

CHEIKH SEYIDI MOHAMED JEDOU

AI Engineer | Machine Learning | Computer Vision | Multi-Agent Systems

Personal Details:

Phone: +222 37 88 08 85 | **Email:** chkseyidi@gmail.com | **Address:** Nouakchott, Mauritania

Education

- 2022 – 2024**
Dakar, Senegal
Master's Degree in Artificial Intelligence and Big Data
Cheikh Anta Diop University, ESP.
- 2019 – 2022**
Nouakchott, Mauritania
Bachelor's Degree in Mathematics and Computer Science
University of Nouakchott Al Asriya, FST.

Professional Experience

- 2025 – 2026**
Nouakchott
AI Engineer – Smart MS SA:
Architected and shipped production AI systems spanning Machine Learning, Computer Vision, and NLP for low-resource environments — combining custom embeddings, vision pipelines, and multi-agent orchestration. Owned the full model lifecycle from training through deployment in environments where pre-trained models don't exist.
- 2024**
Nouakchott
AI Intern – GARini Project, Ministry of Digital Transformation:
Developed GARini, an intelligent educational assistant built with Django. Designed semantic search pipelines with FAISS vector indexing, custom embeddings, and contextual reasoning to deliver accurate, domain-specific educational responses. Implemented response evaluation and quality control mechanisms for agentic workflows.
- 2022**
Nouakchott
Research Data Engineering – PEV Sahel Project, G5 Sahel:
Research data engineering for a G5 Sahel / Coventry University field investigation, in collaboration with the University of Nouakchott Al Asriya. Built pipelines for collection, cleaning, validation, and analytical visualization — supporting investigator training and ensuring data integrity across the study.
- Current**
Remote
Freelance – Autonomous AI Agents & Multi-Agent Systems:
Designing autonomous agents and multi-agent systems: custom orchestration logic, multi-step planning, tool use and function calling, persistent memory, and execution loops with error recovery and self-correction. Selective freelance engagements on hard agent problems.

Technical Skills

Machine Learning

- PyTorch
- Scikit-learn
- XGBoost & LightGBM
- Feature engineering
- Transformer architectures
- Transfer learning

Computer Vision

- OpenCV
- YOLO / Ultralytics
- Image segmentation
- Vision Transformers (ViT)
- Data augmentation
- Transfer learning (vision)

NLP & LLMs

- PEFT / LoRA fine-tuning
- Custom embeddings
- Tokenizer design
- Low-resource NLP
- Model evaluation
- Quantization & distillation

Agents & Orchestration

- Multi-agent systems
- Tool use & function calling
- Multi-step planning
- Custom orchestration
- Persistent memory
- Self-correction loops

Semantic Search & Retrieval

- FAISS · Qdrant
- Vector indexing
- Semantic search
- Contextual reasoning
- Hybrid retrieval
- Domain-specific pipelines

Engineering & MLOps

- Python
- FastAPI · Django
- Docker
- Hugging Face ecosystem
- ML pipelines
- Production deployment

Languages

• **English**
Advanced

• **Français**
Advanced

• **Arabic**
Advanced